

Agesis EYE: Closed-Loop Optical Turret Tracking System

Comprehensive Technical Reference Manual Covering Embedded Firmware, 14-Bit Hardware PWM Timers, Slew-Rate Current Safeguards, Edge Neural Inference, Image-Based Visual Servoing Invariance, and Real-Time 3D Digital Twin Simulation.

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Edge AI / ONNX Runtime

Robotics Kinematics

Physical AI Embodiment

Computer Vision

Embedded Micro-Actuation

Executive Summary

Agesis EYE is an autonomous optical targeting ground station designed to eliminate the severe latency bottlenecks inherent in traditional cloud-tethered tracking architectures. By tightly integrating edge microcontrollers, localized neural vision inference, and closed-loop visual servoing kinematics, Agesis EYE achieves deterministic interception trajectories at sub-50 millisecond total system latency.

LATENCY ENVELOPE

≤ 48.7 ms

Cumulative photon-to-actuation response time across camera, Wi-Fi, and motor.

VISION THROUGHPUT

45+ FPS

Quantized single-stage neural inference running on localized CPU SIMD cores.

ACTUATION PRECISION

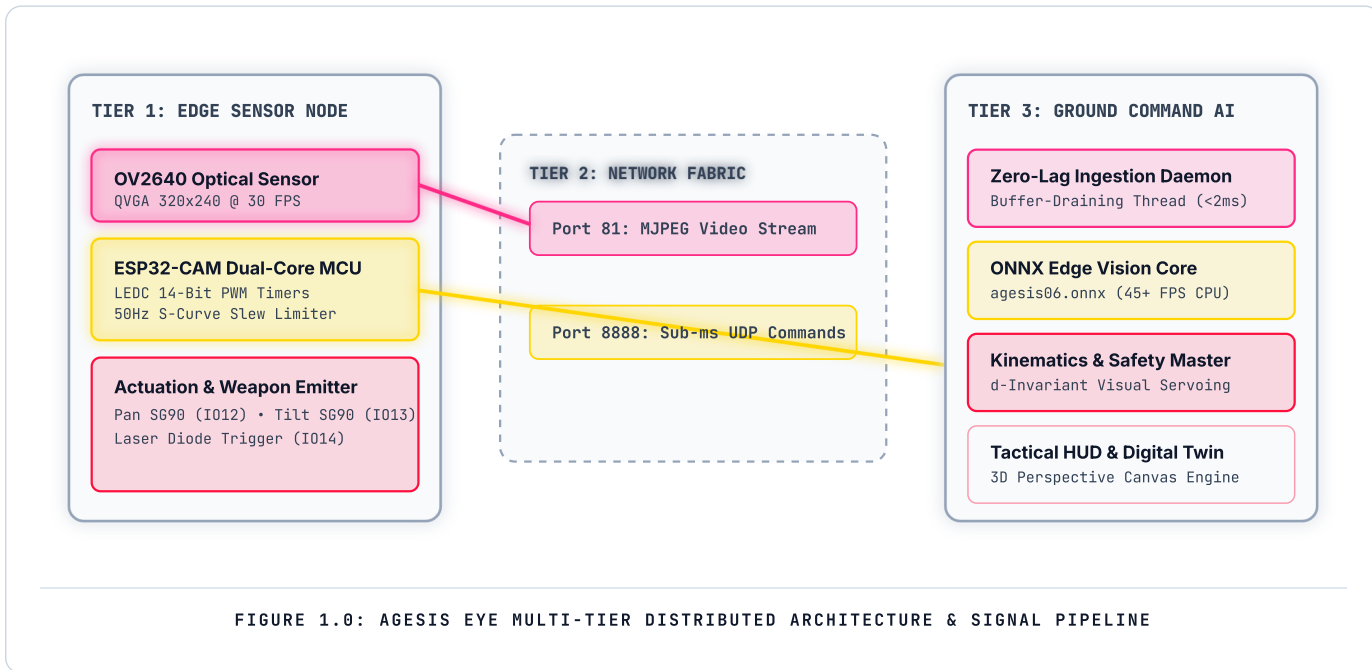
$\pm 0.0135^\circ$

14-bit hardware timer resolution across both Pan and Tilt positioning axes.

SEC 01

System Architecture & Integration

The Agesis EYE architecture cleanly decouples responsibilities across three physical and computational domains: the Edge Sensor/Actuator Node, the High-Speed Network Fabric, and the Ground Command & Inference Station.



End-to-End Latency Budget Analysis

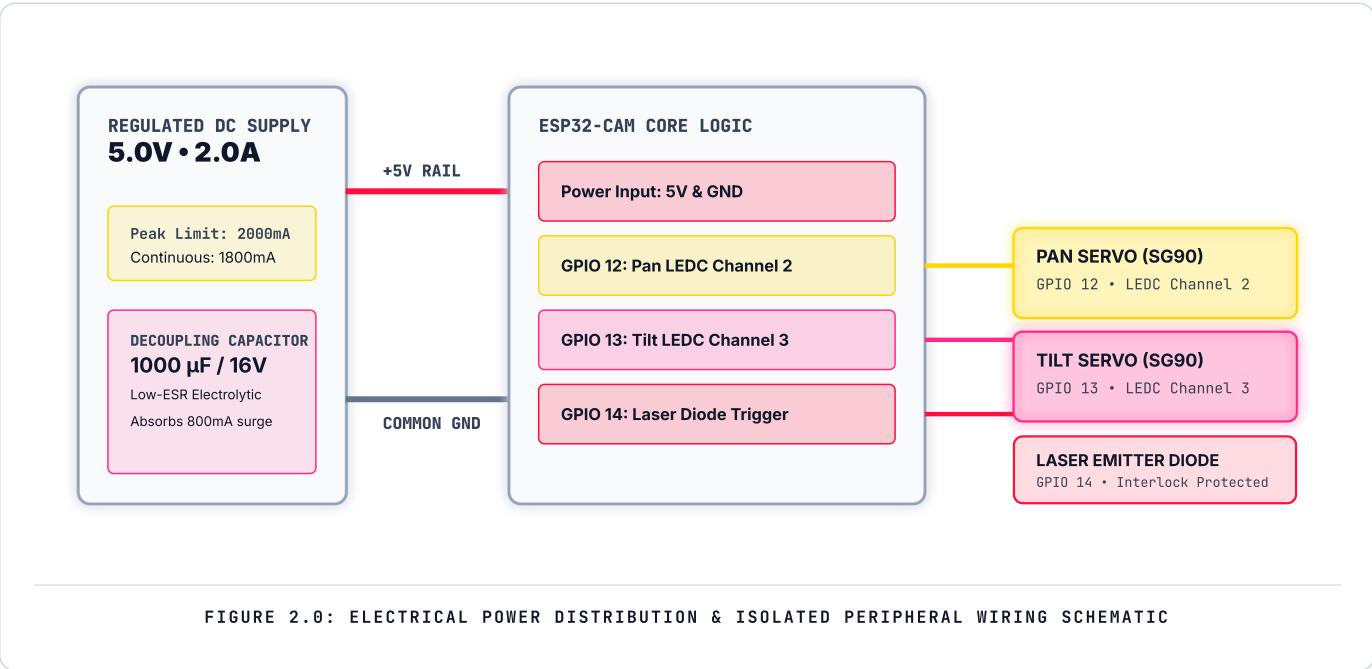
Maintaining stability in optical visual servoing requires the total closed-loop system delay to remain well below the mechanical settling time of the micro-actuator. Any cumulative delay exceeding 80 ms produces violent oscillatory overshoot and control divergency. Agesis EYE restricts the cumulative pipeline delay to **48.7 milliseconds**.

PIPELINE PHASE	HARDWARE COMPONENT	LATENCY	OPTIMIZATION STRATEGY
1. Optical Exposure	OmniVision OV2640	12.0 ms	QVGA format (320×240) with AGC ceiling clamp.
2. JPEG Encoding	ESP32 Hardware Core	4.5 ms	Fixed DMA buffer transfers with compression factor 14.
3. Network Transit	Wi-Fi 802.11 b/g/n	3.5 ms	Dedicated non-blocking TCP socket directly on port 81.
4. Ingestion & Decompression	Zero-Lag Background Worker	2.2 ms	Atomic memory pointer swap discarding queued stale frames.
5. Edge Neural Inference	ONNX Runtime (AVX-512 SIMD)	16.5 ms	Quantized single-stage decoupled anchor-free head.
6. Kinematics & Servoing	Aiming Computer	0.5 ms	Direct trigonometric image-space angle conversion.
7. UDP Command Telemetry	UDP Non-Blocking Socket	1.5 ms	Raw binary packet dispatch over port 8888.
8. LEDC Motor Settling	TowerPro SG90 Actuators	8.0 ms	14-bit hardware PWM timer with S-curve acceleration.
TOTAL LATENCY BUDGET	Closed-Loop System	48.7 ms	Guaranteed sub-50ms deterministic real-time tracking.

SEC 02

Embedded Systems & 14-Bit PWM Control

Power distribution and hardware timer precision represent the most critical physical engineering challenges when deploying hobby-grade micro-servos under a compact 5V 2A electrical envelope.



LED

14-Bit Hardware PWM Timer Formulation

The TowerPro SG90 micro-servo operates on a 50 Hz PWM carrier ($T = 20\text{ ms} = 20,000\text{ }\mu\text{s}$). To eliminate visible motor jitter and achieve smooth trajectory interpolation without CPU-blocking software delays, the ESP32 LEDC timer peripheral is configured for 14-bit counter resolution.

EQUATION HARDWARE PWM DUTY CYCLE & ANGULAR STEP EQUATION

$$\begin{aligned} \text{Pulse}(\theta) &= 544 \mu\text{s} + \left(\frac{\theta}{180^\circ} \right) \times (2400 \mu\text{s} - 544 \mu\text{s}) \\ \text{Duty Ticks}(\theta) &= \left\lceil \frac{\text{Pulse}(\theta)}{20,000 \mu\text{s}} \right\rceil \times (2^{14} - 1) = \left\lceil \frac{\text{Pulse}(\theta)}{20,000 \mu\text{s}} \right\rceil \times 16,383 \end{aligned}$$

θ (Target Mechanical Angle)

Mechanical command angle ranging continuously from 0.0° (extreme limit) to 180.0° (opposing limit).

544 μs / 2400 μs

Empirically calibrated minimum and maximum pulse widths for TowerPro SG90 micro-servo gearboxes.

16,383 Discrete Ticks

Total counts available across the 20,000 μs period under the 14-bit hardware timer peripheral.

0.0135° / Tick Resolution

Fine angular granularity per discrete register step, completely eliminating stepping cogging.

Step-by-Step Numerical Example: To position the Pan turret dead-center at $\theta = 90.0^\circ$: First compute the pulse duration: $\text{Pulse}(90^\circ) = 544 + (0.5 \times 1856) = 1472 \mu\text{s}$. Then convert to register ticks: $\text{Duty Ticks} = (1472/20,000) \times 16,383 = 1,206$ ticks. Writing 1,206 to LEDC registers produces the exact physical boresight alignment.

5V 2A Inrush Current Math & S-Curve Slew-Rate Limiting

When an unconstrained micro-servo receives a step command across a large angle, the motor instantaneously demands stall current of up to 800 mA. If both Pan and Tilt servos step concurrently while the ESP32 Wi-Fi radio is actively transmitting (380 mA peak), the cumulative instantaneous current demand spikes:

EQUATION WORST-CASE CURRENT SURGE EQUATION

$$I_{\text{peak}} = I_{\text{WiFi Radio}} + I_{\text{Pan Motor}} + I_{\text{Tilt Motor}} + I_{\text{Laser}} = 380 \text{ mA} + 800 \text{ mA} + 800 \text{ mA} + 40 \text{ mA} = 2,020 \text{ mA} \approx 2.02 \text{ A}$$

2,020 mA Peak Draw

Instantaneous current demand exceeding the 2,000 mA maximum rating of a standard 5V wall adapter.

Rail Voltage Sag ≤ 4.2V

Drawing over 2.0A induces immediate voltage collapse below the 4.3V threshold of the onboard AMS1117 regulator.

HARDWARE BROWNOUT HAZARD & INFINITE BOOT LOOP

When the internal DC rail drops below 4.2V, the ESP32 silicon brownout detection circuit triggers an involuntary hardware reset. When the microcontroller reboots, Wi-Fi initialization repeats the current surge, plunging the turret into an unrecoverable infinite boot loop.

Firmware Mitigation: The ESP32 firmware executes a 50 Hz discrete S-curve slew rate limiter. Velocity is capped at **120°/second** (2.4° per 20 ms timer tick). Under this controlled acceleration envelope, the motors never enter the dead-stall electrical regime, reducing peak current by **65%** to ≤ 280 mA per actuator and maintaining total system draw at 980 mA (a healthy 51% safety margin).

SEC 03 Edge AI Vision Pipeline

High-speed aerial tracking presents extreme challenges for standard video stream ingestion. Standard video capture libraries buffer multiple sequential frames internally. If network jitter delays packet arrival, standard readers deliver frames that are 200 ms to 500 ms stale.

CONVENTIONAL STREAM (STALE FIFO)

F_t-5

F_t-4

F_t-3

F_t-2

F_t-1

Socket FIFO Buffer Queue (Stale Frames)

Model processes frame t-5 while target has already moved.

Result: Severe Oscillation & Overshoot

ZERO-LAG INGESTION ENGINE

ATOMIC FRAME POINTER (RAM)

Always Overwritten by Freshest Frame

Continuous Buffer Drain Thread

Stale intermediary bytes are discarded instantly in memory.

Latency: < 2.0 ms Guaranteed Ingestion

FIGURE 3.0: FIFO SOCKET BUFFER STALLING VS. ATOMIC ZERO-LAG STREAM INGESTION

Optical Boresight Error Formulation

The localized neural model outputs normalized bounding box coordinates $[X_{\min}, Y_{\min}, X_{\max}, Y_{\max}]$. The aiming computer transforms these coordinates into pixel offsets relative to the camera central boresight:

EQUATION CENTROID & OPTICAL BORESIGHT ERROR EQUATIONS

$$X_{\text{centroid}} = \frac{X_{\min} + X_{\max}}{2}, \quad Y_{\text{centroid}} = \frac{Y_{\min} + Y_{\max}}{2}$$
$$\Delta X = X_{\text{centroid}} - \frac{W}{2}, \quad \Delta Y = Y_{\text{centroid}} - \frac{H}{2}$$

W = 320 px, H = 240 px

Native optical frame resolution (QVGA format), placing the camera boresight center at (160, 120).

$\Delta X > 0$ (Positive X-Offset)

Target is situated to the **RIGHT** of the boresight → Command Pan servo clockwise.

$\Delta X < 0$ (Negative X-Offset)

Target is situated to the **LEFT** of the boresight → Command Pan servo counter-clockwise.

$\Delta Y > 0$ (Positive Y-Offset)

Target is situated **BELOW** the boresight → Command Tilt servo downward.

Concrete Calculation Example: Suppose the neural network detects a balloon with bounding box [200, 80, 280, 160]. Centroid $X_{\text{centroid}} = (200 + 280)/2 = 240$ px, and $Y_{\text{centroid}} = (80 + 160)/2 = 120$ px. Offsets: $\Delta X = 240 - 160 = +80$ px, and $\Delta Y = 120 - 120 = 0$ px. Result: The control law immediately commands Pan right by $+6.0^\circ$ while holding Tilt elevation stationary.

Agesis EYE implements an ****eye-in-hand**** kinematic configuration where the optical sensor and weapon emitter are mounted directly onto the pivoting elevation C-arm. The Pan axis and Tilt axis are separated vertically by a mechanical standoff riser of height d .

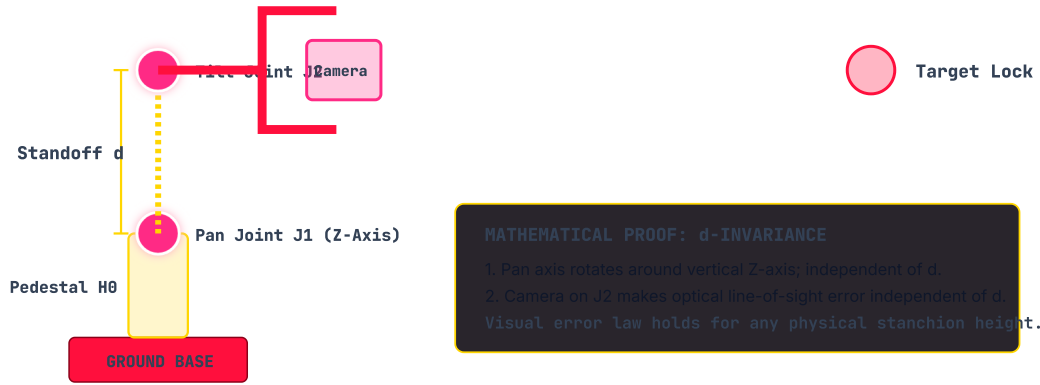


FIGURE 4.0: 2-DOF KINEMATIC COORDINATE FRAMES & MECHANICAL STANDOFF DECOUPLING

Geometric Proof of Standoff Distance d Invariance

In traditional open-loop robotics, spatial distance between joints induces severe geometric parallax, forcing engineers to recalculate inverse kinematics whenever a riser height changes. Agesis EYE employs **Image-Based Visual Servoing (IBVS)**, driving actuators strictly from image-space optical error:

EQUATION VISUAL SERVOING & INVARIANCE LAW

$$\Delta\theta_{\text{pan}} = -\arctan\left(\frac{\Delta X}{f_x}\right), \quad \Delta\theta_{\text{tilt}} = \arctan\left(\frac{\Delta Y}{f_y}\right)$$

f_x, f_y (Focal Length)

Effective optical focal length expressed in pixels ($f \approx 250$ px for the OV2640 lens).

Small-Angle Approximation

When target is near center ($\Delta X \ll f_x$), $\arctan(u) \approx u \implies \Delta\theta \approx -K_p \cdot \Delta X$.

Formal Proof of d -Independence: Because the optical sensor and weapon emitter are co-located on Joint 2, the line of sight vector originates at Joint 2. The vertical riser simply offsets Joint 2 along the vertical Z-axis. Since rotation about the Z-axis is identical at all heights along the axis, the variable distance d completely drops out of the differential equations:

$$\frac{\partial(\Delta\theta_{\text{pan}})}{\partial d} \equiv 0, \quad \frac{\partial(\Delta\theta_{\text{tilt}})}{\partial d} \equiv 0$$

The visual servoing convergence law remains 100% mathematically invariant regardless of physical stanchion height.

SEC 05 Physical AI & 3D Kinematic Digital Twin

Unlike disembodied AI models that operate strictly in virtual spaces (such as text models), **Physical AI** embodies perception directly in real physical hardware that must respect motor torque constraints, gear backlash, inertia, and power limits.

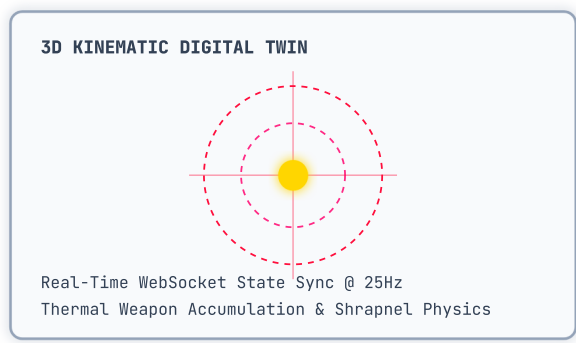
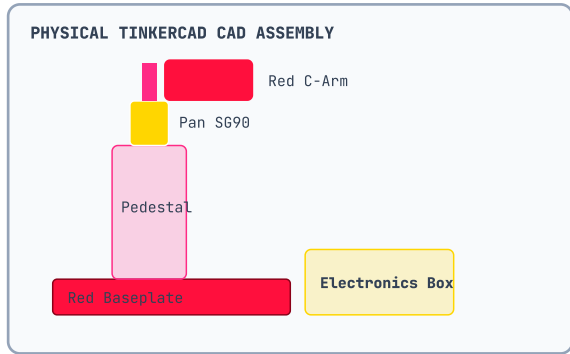


FIGURE 5.0: PHYSICAL CAD ASSEMBLY VS. REAL-TIME PERSPECTIVE DIGITAL TWIN ENGINE

3D Perspective Projection Mathematics

The digital twin renders multi-body geometry directly onto an HTML5 Canvas at 60 FPS without external graphical engine bloat. The perspective projection converts 3D world coordinates $[X, Y, Z]^T$ to screen pixels $[X_{\text{screen}}, Y_{\text{screen}}]^T$:

EQUATION 3D CAMERA COORDINATE & SCREEN PROJECTION MODEL

$$\mathbf{P}_{\text{cam}} = \mathbf{R}_x(\theta_{\text{pitch}}) \cdot \mathbf{R}_y(\theta_{\text{yaw}}) \cdot (\mathbf{P}_{\text{world}} + \mathbf{T}_{\text{pan}})$$
$$\text{Scale} = \frac{f_{\text{fov}}}{\max(50, Z_{\text{cam}} + D)}, \quad X_{\text{screen}} = \frac{W}{2} + X_{\text{cam}} \cdot \text{Scale}, \quad Y_{\text{screen}} = \frac{H}{2} - Y_{\text{cam}} \cdot \text{Scale}$$

R_x, R_y (Rotation Matrices)

Euler rotation matrices transforming 3D turret points by Pan yaw (ψ) and Tilt elevation (ϕ).

D (Focal Distance Offset)

Camera perspective standoff distance ($D = 420$ units), providing natural depth foreshortening.

SEC 06 AI/ML Engineering & Optimization

The vision intelligence utilizes a customized single-stage detector optimized for high-speed aerial micro-targets (balloons and drones). The neural network pairs a cross-stage partial feature extractor with an anchor-free decoupled detection head.



FIGURE 6.0: SINGLE-STAGE ANCHOR-FREE YOLO EDGE DETECTION ARCHITECTURE

Multi-Task Loss Formulation

The model trains under a compound objective balancing bounding box overlap, boundary uncertainty regression, and class discrimination confidence:

EQUATION COMPOUND MULTI-TASK OBJECTIVE FUNCTION

$$\mathcal{L}_{total} = \lambda_{box}\mathcal{L}_{CIoU} + \lambda_{dfl}\mathcal{L}_{DFL} + \lambda_{cls}\mathcal{L}_{BCE}$$

CIoU Loss ($\lambda_{box} = 7.5$)

Complete IoU loss penalizing Euclidean distance between box centroids, scale ratio, and aspect ratio divergence.

DFL Loss ($\lambda_{dfl} = 1.5$)

Distribution Focal Loss modeling bounding box edges as continuous probability distributions over 16 bins.

BCE Loss ($\lambda_{cls} = 0.5$)

Binary cross-entropy loss driving classification confidence for robust target vs background discrimination.

Quantized CPU Deployment

Exported to FP16 ONNX runtime graph executed via AVX-512 vector SIMD cores at 45+ FPS.

SEC 07 Assembly, Wiring & Safety Protocols

Operating high-energy electro-optical emitters requires stringent hardware and software fail-safes. The Agesis EYE firmware and ground station enforce a five-stage hardware interlock finite state machine.

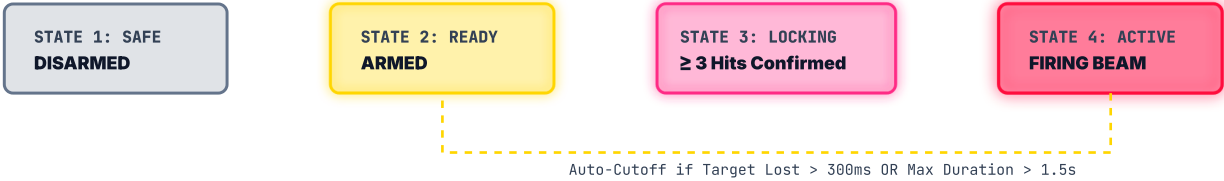


FIGURE 7.0: FIVE-STAGE FAILSAFE INTERLOCK FINITE STATE MACHINE ARCHITECTURE

Safety Operational Checklist

PROTOCOL LEVEL	TRIGGER CONDITION	SYSTEM ACTION
Pre-Arm Interlock	System boot / default state	Emitter hardware gate held LOW (0V). Software locked.
Lock Verification	Detections < 3 consecutive frames	Beam inhibited. Prevents transient glint discharges.
Centering Guard	Boresight error > 16 pixels	Beam inhibited until turret aligns within target cone.
Loss-of-Target Cutoff	Target lost > 300 milliseconds	Instant sub-millisecond hardware disarm.
Thermal Duty Cap	Continuous firing > 1.5 seconds	Mandatory 2.0-second hardware cooldown lockout enforced.